Department of Artificial Intelligence, SVNIT, Surat

B.Tech-III, Semester-V

Subject: Machine Learning (AI301)

LAB ASSIGNMENT

Problem Statement: Use the Abalone dataset (*abalone.csv*) to perform classification.

Part A: Logistic Regression and PCA

1. Read the Abalone dataset (abalone.csv) and inspect its structure.
2. Perform exploratory data analysis (EDA), including suitable summary statistics and visualizations.
3. Prepare the input features and target variable. Encode categorical features where required.
4. Split the dataset into training and testing sets.
5. Standardize the input features using an appropriate feature-scaling method.
6. Build a Logistic Regression model and make predictions on the test dataset.
7. Evaluate the Logistic Regression model using accuracy_score, confusion_matrix, and classification_report.
8. Perform Principal Component Analysis (PCA) on the standardized input features.
9. Print pca_model.explained_variance_ratio_ and interpret the variance explained by the principal components.
10. Visualize the Logistic Regression coefficients associated with the principal components using a heat map.
11. Train and test a Logistic Regression model using the selected principal components.
12. Evaluate the PCA-based Logistic Regression model using accuracy_score, confusion_matrix, and classification_report.

Part B: Support Vector Machine (SVM) Implementation

1. Use the same Abalone dataset and the same target variable defined in Part A.
2. Split the data into training and testing sets. Apply the same preprocessing and standardization procedure; fit the scaler only on the training data.
3. Build an SVM classifier using sklearn.svm.SVC. Begin with a suitable kernel (for example, RBF) and document the chosen parameters.
4. Train the SVM model on the standardized training features and predict the labels for the test set.
5. Evaluate the SVM model using accuracy_score, confusion_matrix, and classification_report.
6. Repeat the SVM experiment using the principal components obtained from PCA in Part A.
7. Compare the SVM results with Logistic Regression, both with the original standardized features and with PCA features. Briefly state which approach performs better on the test set.

Note: Use the same train/test split for fair model comparison. If the target is created from a numeric Abalone measurement (such as Rings), clearly explain how it is converted into classes before classification.